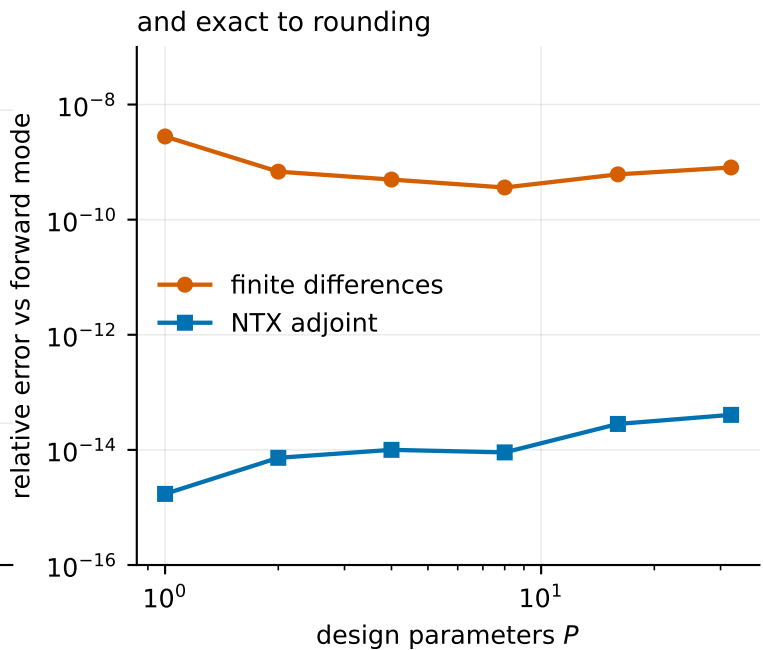
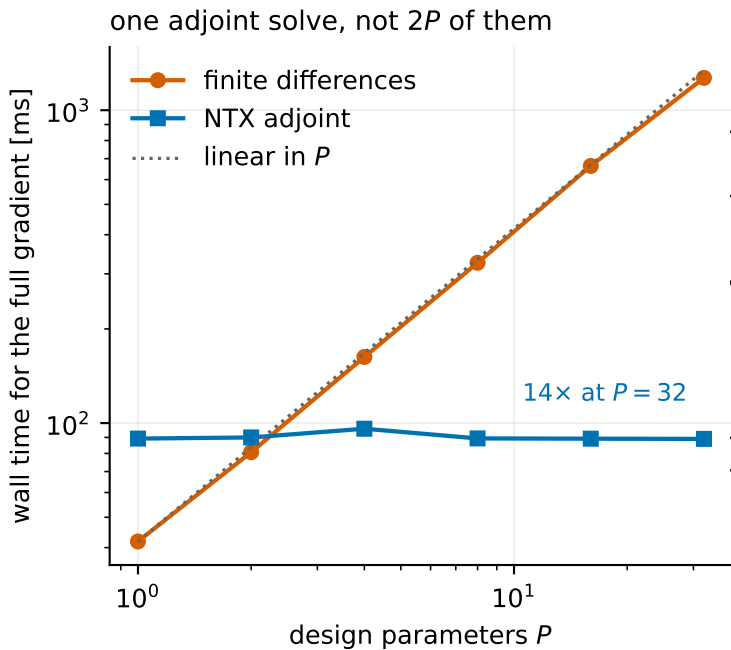




D_{11} on a 13×13 surface grid, $N_\xi = 40$, $\hat{\nu} = 0.01$, float64 on cpu. Finite differences use central steps; forward mode arbitrates.